

Sales Visit Planning and Management Platform

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System Purpose

Actors

Sales Representative

Manager

Administrator

External Mapping / Location Service

Core Workflows

Workflow 1: Day Planning

Workflow 2: Visit and Call Execution

Workflow 3: Expense Submission and Approval

Example Business Rules

Potential Failure Scenarios

Failure Scenario 1: External Location Service Unavailable

Failure Scenario 2: Incorrect Coordinates Returned

Failure Scenario 3: Connectivity Loss Mid-Route

Conclusion

System Purpose

Field-based sales representatives currently manage their working day across disconnected tools, with no centralised system for planning visits, scheduling calls, recording outcomes, or tracking expenses. This results in inefficient routing, poor visibility for management, and no reliable record of daily activity.

The proposed system addresses this by providing a platform for sales representatives to plan and manage their working day, with supporting workflows for managers to oversee activity, approve expenses, and flag priority customers. An administrator role handles user management across the platform.

Actors

Sales Representative

Plans their working day by selecting customers to visit and call, executes the plan in the field, logs visit and call outcomes with notes, submits expenses, and may amend an active plan mid-day if circumstances change.

Manager

Reviews visit logs, call logs, and expense submissions across their assigned representatives.

Approves or rejects expense submissions, flags recommended customers for visits or calls, and manages the active customer pool by deactivating customers where required.

Administrator

Responsible for user management across the platform. Adds and removes Sales Representatives and Managers from the system.

External Mapping / Location Service

Resolves customer addresses into geographic coordinates used by the route optimisation process. Operates outside the system boundary.

Core Workflows

Workflow 1: Day Planning

1. The representative selects customers to visit and call from the active customer pool.
2. Management-recommended customers are highlighted but are not mandatory selections.
3. A start and end location are set, and time blocks assigned to each scheduled call, defaulting to 30 minutes but configurable per call.
4. For any customer without stored coordinates, the system requests location data from the external mapping service and stores the result against the customer record.
5. Visit customers are ordered by a route optimisation process and a mapping service URL is produced for navigation.

Plan states: Draft > Confirmed > Active

Workflow 2: Visit and Call Execution

1. The representative works through their confirmed plan.
2. Visits transition through;
Planned > In Progress > Completed / Missed / Rescheduled
3. Calls transition through;
Scheduled > Attempted > Completed.
4. Outcome and call notes are recorded against each entry.
5. If a customer becomes unavailable mid-day, they can be marked as such and a re-optimisation of the remaining route requested.
6. Completed records are made visible to the representative's manager.

Workflow 3: Expense Submission and Approval

1. The representative logs expenses against their active day plan across permitted categories; **fuel, tolls, meals, and hotel stays.**
2. Submissions transition through;
Submitted > Pending Approval > Approved / Rejected.
3. Rejected expenses may be amended and resubmitted.

Example Business Rules

- A day plan cannot be confirmed without at least one visit or call scheduled.
- A visit cannot be marked as complete without outcome notes being recorded.
- Expense submissions may only be approved or rejected by a Manager.
- If a customer is deactivated while present on a confirmed or active plan, the representative must be notified and the affected stop flagged.
- A representative's historical visit, call, and expense records must be retained if their account is removed from the system.

Potential Failure Scenarios

Failure Scenario 1: External Location Service Unavailable

If the external mapping service is unreachable, the system cannot resolve coordinates for customers without cached location data, leaving the day plan in an incomplete state.

This is an ongoing degraded state rather than a one-off error. The plan cannot proceed until the service recovers, coordinates are manually provided, or the affected customer is removed.

Failure Scenario 2: Incorrect Coordinates Returned

The external service may return inaccurate coordinates for an ambiguous or poorly formatted address. As these are cached back to the database on return, the error silently affects all future route calculations for that customer until identified and corrected.

Failure Scenario 3: Connectivity Loss Mid-Route

A representative may lose connectivity while working through their plan, leaving them unable to log outcomes or expenses. A local device record of the confirmed plan allows continued operation in a degraded state, with changes synchronised on reconnection. Where the server state has changed during the offline period, conflict resolution is required.

Conclusion

This domain provides sufficient complexity for Stage 2 architectural modelling through its multi-actor structure, stateful long-running workflows, dependency on an external service, offline resilience requirements, and cross-actor coordination across planning, execution, and approval processes. The customer lifecycle management aspect introduces additional consistency and coordination challenges that create meaningful design questions beyond data storage and retrieval.